SMART INDIA HACKATHON 2024

TITLE PAGE

- **Problem Statement ID – 1700**
- **Problem Statement Title – Developing an AI based interactive Chatbot or virtual assistant for the Department of Justice's Website.**
- **Theme – Smart Automation**
- **PS Category – Software**
- **Team ID – 1363**
- **Team Name – Code Monkeys**

PROPOSED SOLUTION

SOLUTION →

Development of an AI chatbot that interacts with users to answer their questions related to the Department of Justice (DoJ).

- **Multilingual Support:** The AI chatbot that can answer questions in multiple languages.
- **User-friendly Interface**: The chatbot offers users a simple interface, that can be easily used by citizens, legal representatives, and government staff.
- **Self-learning Capability:** By using machine learning techniques to improve accuracy, the chatbot will get better at answering questions.
- **Integration with DoJ Systems:** Can connects with the National Judicial Data Grid (NJDG), eCourts, Fast Track Courts, Tele Law services, and other relevant judicial platforms for **real-time data** retrieval.

UNIQUENESS →

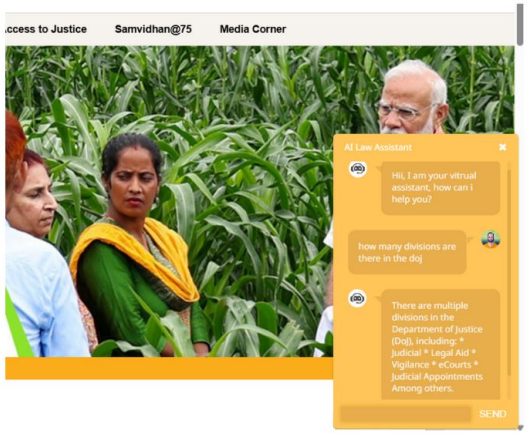
- **AI Model** is developed by our Team by using legal dataset like doj.gov.in and law books.
- **Scalability**: Our solution can scale by increasing the amount of resources (ram, cpu etc) as more services are offered to the users by DoJ.
- **AI That Learns:** Our chatbot gets better with time. It learns from user interactions, which helps it give more correct information.

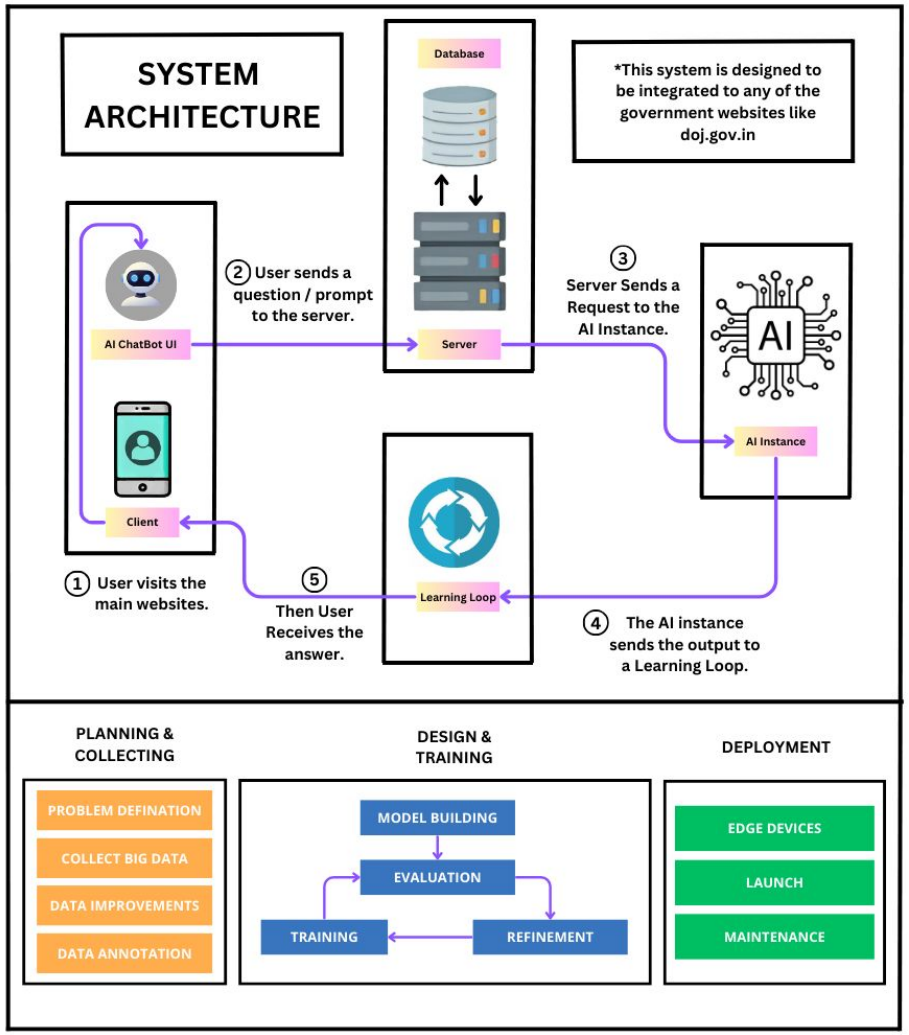
ADDRESSING THE PROBLEM →

The chatbot will help users to avoid going through government web pages to find answers for questions such as court case status, judge appointment and e-filing procedure and reduce the burden on DoJ staffs.

TECHNICAL APPROACH

- **Backend** : Core technologies are **Node JS, Express JS**.
- **Frontend** : **React JS** and **Tailwind CSS** will be used to make beautiful User Interface for the website.
- **AI Model** : Pytorch with legal dataset.
- **APIs** : Google Translation.

Tech Stack			Development in Progress
Front-End	Back-End	Others	
React JS	Node JS	Pytorch	
Tailwind	Express JS	Google Translation	
Bootstrap	Python for AI		



FEASIBILITY AND VIABILITY

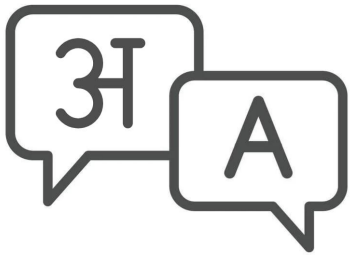
- **Analysis of Feasibility →**
  - The tech stack can scale efficiently by adding resources or more servers to handle more users and data.
  - The cost of hosting the web application:-
    - Google Cloud E2 Medium Instance - **₹10,000**/month.
    - Our AI model - **₹0** (It will run locally).
    - Google Translation API - **₹5,000**/month.

- **Potential Risks & Challenges →**
  - **Data Availability:** High-quality training data in regional languages might be limited compared to English and Hindi.
  - **User Adoption:** Public unfamiliarity with digital judicial services might slow adoption.
- **Strategies For Overcoming These Challenges →**
  - Starting with major languages such as Hindi and Tamil, gradually adding more languages and use **Google Translation API** for less widely spoken languages.

Cost Comparision

ChatGPT	+	Translation API	=	₹60,000 / mo
Gemini	+	Translation API	=	₹50,000 / mo
OUR-AI	+	Translation API	=	₹15,000 / mo
Vertex AI	+	Translation API	=	₹35,000 / mo

*More details on research page



IMPACT AND BENEFITS

IMPACT ON THE TARGET AUDIENCE →

- **Increased Trust in the Legal System:** Real-time data on court proceedings increases transparency and builds public trust.
- **Easier Access to Legal Information:** Citizens can quickly get legal information by simply clicking on the chatbot and asking a question in their language.

ECONOMIC BENEFITS →

- **Cost Savings for DoJ:** Automating responses to regular queries can cut down the need for human resources.
- **Economic Accessibility:** Offering a free and simple platform helps citizens from various economic backgrounds access the legal services without a huge financial burden.

SOCIAL BENEFITS →

- **Inclusivity:** Since the chatbot is multilingual, it bridges the language gaps which makes the justice system more accessible for non-English speakers.
- **Timely Assistance:** By letting citizens access legal information the chatbot empowers users to take prompt action in legal matters.

ENVIRONMENTAL BENEFITS →

- **Paperless Transactions:** Encouraging the use of digital services such as e Filing reduces the usage of paper decreasing environmental waste.
- **Reduced Carbon Footprint:** With access to legal services, citizens will not have to travel to courts which will reduce transportation related emissions improving environmental sustainability.

RESEARCH AND REFERENCES

- **AI Model research →**
  - Fine Tuning the AI Model using pytorch and pretrained transformers.
  - https://pytorch.org/blog/finetune-llms/
- **Cost Comparison →**
  - https://drive.google.com/file/d/1LLBtgHimlXVUyjpfESrEeoc6QCTW6kZm/view?usp=drive_link
- **Training Dataset →**
  - https://njdg.ecourts.gov.in/njdgnew/index.php
  - https://doj.gov.in/
  - https://www.msrlawbooks.in/
  - dataset in prototype : https://drive.google.com/file/d/...
- **Working Prototype Demo →**
  - https://youtu.be/UXvvrmlAqxc